

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DevOps Lab Experiment – 9

Introduction to Azure DevOps: Overview of Azure DevOps Services, Setting Up an Azure DevOps Account and Project

Overview of Azure DevOps Services

Azure DevOps is a **cloud-based platform** that provides integrated tools for software development teams. It includes everything you need to plan work, collaborate on code, build applications, test functionality, and deploy to production.

Azure DevOps includes the following integrated services

Azure Boards: Plan and track work using Agile tools, Kanban boards, backlogs, and dashboards. Create work items like user stories, bugs, and tasks. Use sprint planning, burndown charts, and velocity tracking. Customize workflows and work item types to match your team's process.

Example scenario: A product team planning a mobile app feature creates user stories for "user sign-in," tracks bugs found during development, and uses sprint boards to monitor progress during two-week iterations.

Azure Repos: Host unlimited private Git repositories or use Team Foundation Version Control (TFVC) for source code management. Features include branch policies, pull requests with code reviews, conflict resolution, and integration with popular IDEs and editors.

Example scenario : Development team members create feature branches for new functionality, submit pull requests for code review, and use branch policies to ensure all code is reviewed and tested before merging to the main branch.

Azure Pipelines: Build, test, and deploy applications with CI/CD pipelines that work with any language, platform, and cloud. Supports Docker containers, Kubernetes, and deployments to Azure, AWS, Google Cloud, or on premises. Includes parallel jobs, deployment gates, and release approvals.

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Example scenario : Every code commit triggers an automated pipeline that builds a .NET web application, runs unit tests, creates a Docker container, and deploys to staging environment for testing before production release.

Azure Test Plans : Plan, execute, and track testing with manual test cases, exploratory testing sessions, and automated test integration. Create test suites, track test results, capture screenshots and videos, and generate detailed test reports.

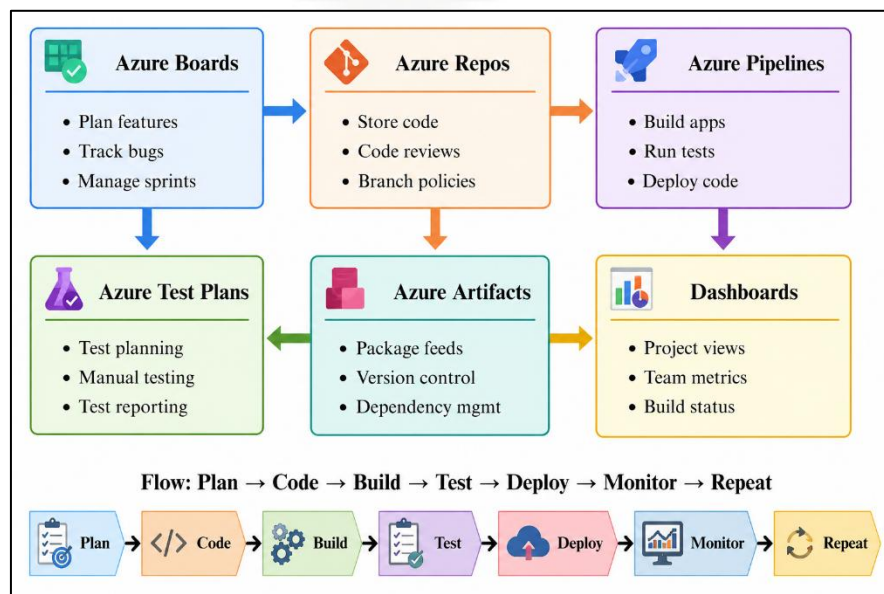
Example scenario : QA team creates test cases for user registration flow, executes manual tests on different browsers, captures screenshots of issues, and links test results to user stories for traceability.

Azure Artifacts : Create, host, and share packages like NuGet, npm, Maven, Python, and Universal packages with your team and organization. Integrate with build pipelines, manage package versions, and control access with upstream sources and retention policies.

Example scenario : Development team creates a shared authentication library, publishes it as a NuGet package to Azure Artifacts, and references it across multiple projects while controlling access to internal packages.

How Azure DevOps services work together

The following diagram shows how the services integrate throughout the development lifecycle.



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Typical workflow:

- Plan** work items in Azure Boards
- Code** features in Azure Repos with pull requests
- Build** and package with Azure Pipelines and Azure Artifacts
- Test** manually and automatically using Azure Test Plans
- Deploy** through Azure Pipelines to various environments
- Monitor** progress and metrics via Dashboards
- Iterate** based on feedback and new requirements

Setting Up an Azure DevOps Account

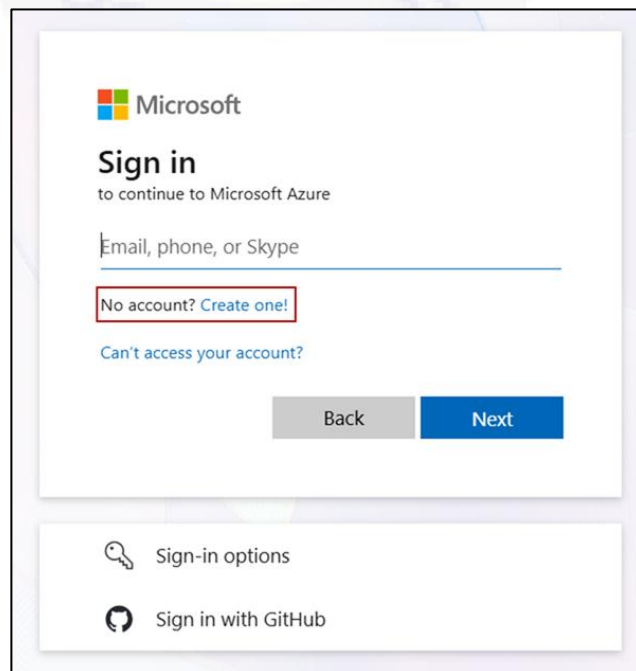
Before using Azure DevOps, you must create an account and set up your organization

Step 1: Sign Up for an Azure DevOps Account

- Open a browser and navigate to <https://azure.microsoft.com>
- **Sign in** using a **Microsoft email account** (Outlook / Hotmail / Office 365 / Microsoft 365)

OR

If you do not have a Microsoft account, click **Create one!** to create a new **Microsoft email account** and follow the instructions.

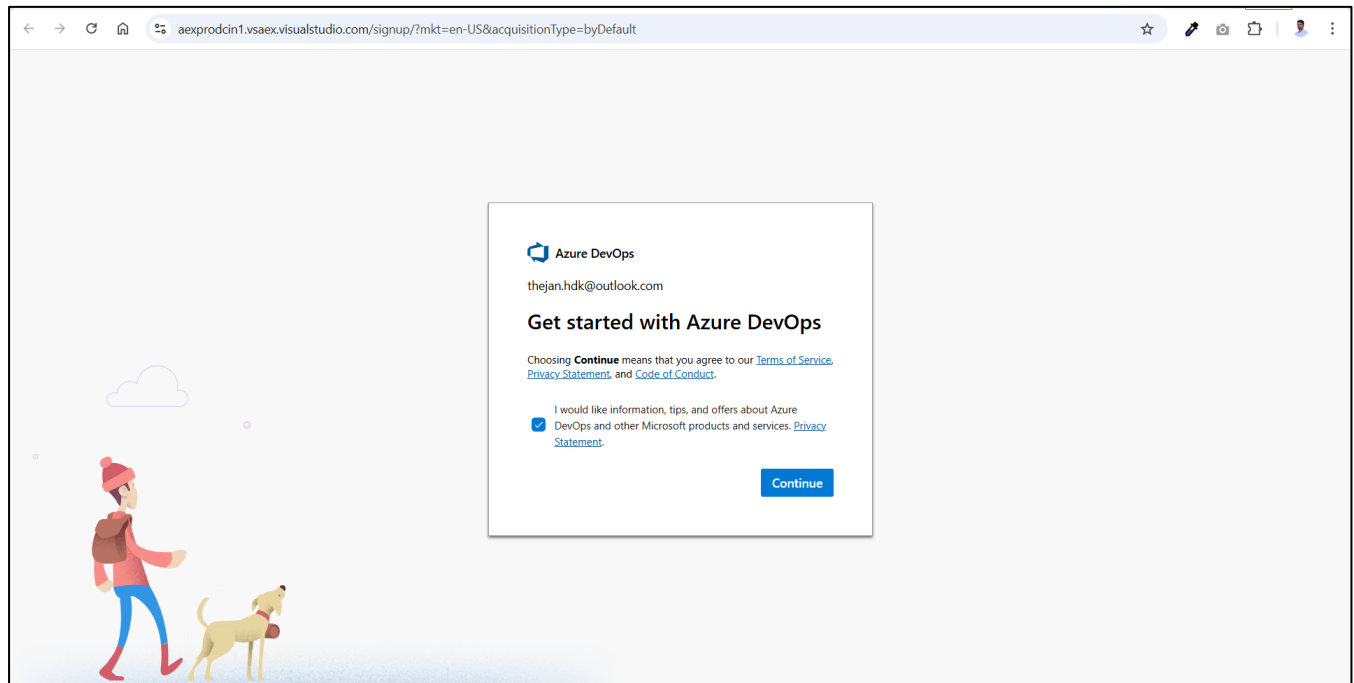


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- If prompted, review and accept Microsoft's Terms of Service to continue.

Create an Azure DevOps Organization

- After signing in with your Microsoft account, the **Get started with Azure DevOps** page appears.
- Click **Continue** to proceed with the Azure DevOps setup

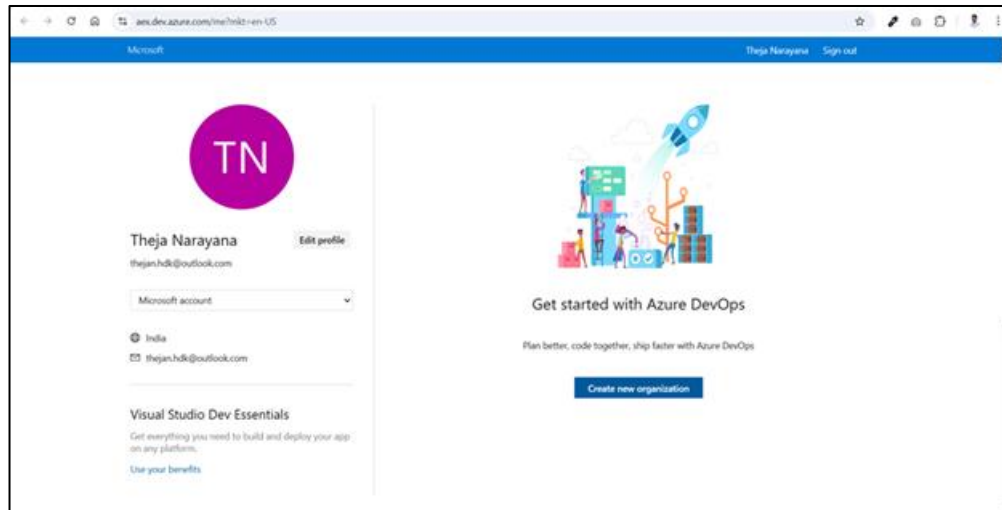


The initial setup (welcome) page for Azure DevOps and it appears the first time you sign in to Azure DevOps with your Microsoft account.

When you click on **Continue**, you will be prompted to create an Azure DevOps organization and if not prompted, Click on **Menu → DevOps → Create new organization**

- Enter a unique name for the organization
- Select a country where your project will be hosted
- Click on **Get started with Azure** to link your organization to **Azure DevOps Subscription**
- Click **Continue**

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Azure DevOps

thejan.hdk@outlook.com [Switch directory](#)

Almost done...

Name your Azure DevOps organization *

1

We'll host your projects in

2

Select an Azure subscription for billing [12](#)

3

You can also [create a new Azure subscription](#)

1 This subscription has a spending limit. Any paid usage will disable your subscription to protect you from accidentally incurring charges. [Learn more about free tier limits](#)

Enter the characters you see

New Audio



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Setting Up an Azure DevOps Project

- Go to your Azure DevOps organization dashboard.
- Click the + **New project button** located at the top-right corner of the page.
- In the Create a new project window, enter the following details:
 - Project name (e.g., *DevOpsExperiment-9*)
 - Description
- Select the Visibility:
 - Private (recommended for class/lab work)
 - Public (anyone can view the project)
- Select Version control → Git.
- Select Work item process → Agile (default).
- Click **Create**.



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Create new project ×

Project name *

Description

Visibility

Private

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ Advanced

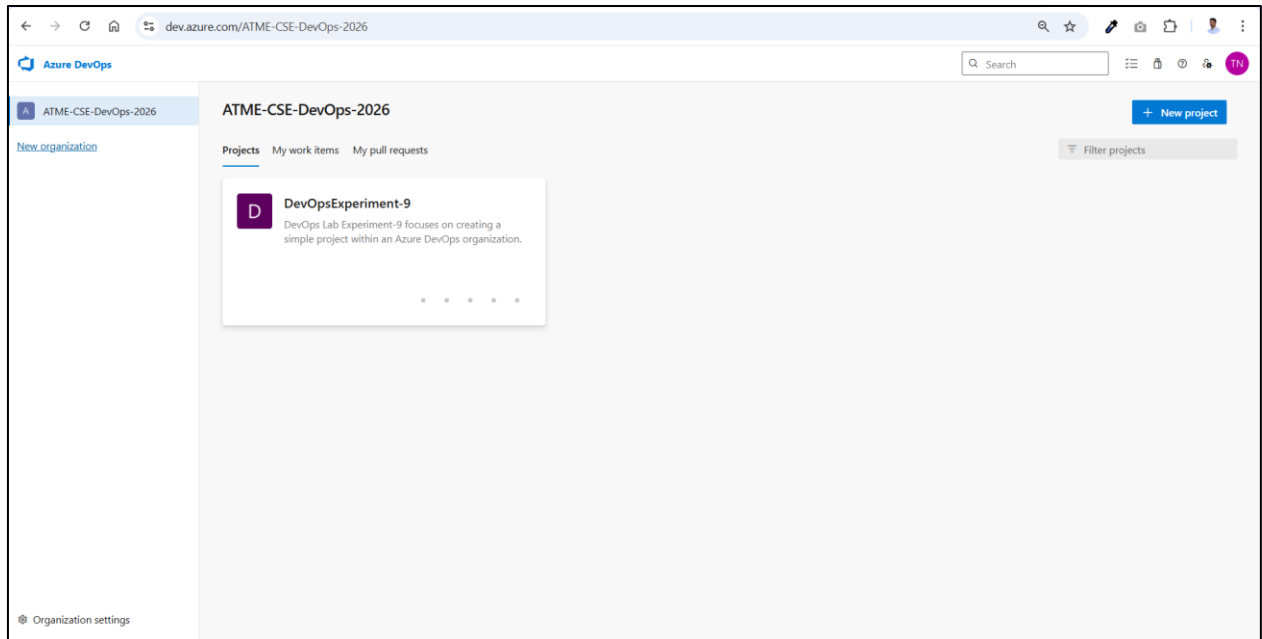
Version control ?

Work item process ?

Cancel

Create

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VIVA Questions

1. What is Azure DevOps?
2. What are the main services provided by Azure DevOps?
3. What is the purpose of Azure DevOps in software development?
4. What is an Azure DevOps organization?
5. What is the difference between Azure and Azure DevOps?
6. Explain the components of Azure DevOps such as Repos, Pipelines, Boards, Test Plans, and Artifacts.
7. What is version control and why is Git used in Azure DevOps?
8. What is the difference between Git and centralized version control systems?
9. What is a project in Azure DevOps?
10. What is meant by project visibility (Public vs Private)?
11. What is the Agile process in Azure DevOps?
12. What are the steps to create an Azure DevOps account?
13. How do you create an Azure DevOps organization?
14. What details are required while creating a new project?

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15. Why is Git selected as version control while creating a project?
16. How do you access the Azure DevOps organization dashboard?
17. What is the role of Azure DevOps in CI/CD?
18. How does Azure DevOps support team collaboration?
19. Can Azure DevOps be integrated with GitHub? How?
20. What are the advantages of using cloud-based DevOps tools?
21. What issues can arise if an incorrect region is selected while creating an organization?
22. Compare Azure DevOps with Jenkins.
23. What security features are available in Azure DevOps?
24. How does Azure DevOps support scalability?
25. Define CI/CD.
26. What is a repository?
27. What is a pipeline?
28. What is a work item in Azure DevOps?
29. What is Agile methodology?